

JANUS Architecture

Physics-Level Error Analysis, Corrections & Verified Solutions

Patent Application No. 202611052791 · Deepanshu Bhardwaj

Prepared from cross-document analysis of Janus.pdf and Refinement_lv1.txt

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■ Abstract

This document provides a rigorous physics-level audit of the JANUS hybrid photonic-electronic compute architecture (Patent Application No. 202611052791). Five verified errors are identified across the timing model, GaP microring spectral matching, optical power budget, laser source specification, and CMOS backend design. Each error is characterised with the original paper's claim, the contradiction, the quantitative derivation of the correct value, and a verified engineering fix with references to demonstrated industry solutions. A corrected architecture summary and open-problem registry complete the document.

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1. Executive Summary

What was wrong, what was fixed, and what the net architecture looks like

Scope of Analysis

The analysis covers five distinct physics-level errors identified by cross-referencing the full architectural specification (Janus.pdf, Patent 202611052791) against an independent stress-test session (Refinement_lv1.txt). All five errors are verified independently below with first-principles derivations. None of the errors affect the core architectural innovations—One-Hot spatial encoding, PCM Beneš routing, CRT fault visibility, or Z-axis thermal partitioning—which remain physically valid.

Error Inventory at a Glance

Error	Section	Nature	Impact	Status
E1: Timing model	IV.J.1 Eq.247–250	Serial sum applied to pipelined system	33→100 GHz, 3× throughput gain	Fixed: wave-pipeline model
E2: GaP Q-factor	III.D.4	$Q > 10,000$ with 1.5 ps pulses — 15× spectral mismatch	Power loss, broken timing	Fixed: $Q \approx 2,000$, 5 ps pulses
E3: Detector power	III.B.5 narrative	−7.5 dBm stated vs +2.5 dBm actual	Margin understated by 10 dB	Fixed: narrative correction
E4: Laser power	III.E.3, Table XVII	1.6 mW avg vs 0.032 mW correct; Table XVII ×50 too large	Laser budget overstated 50×	Fixed: 0.44 W datacenter
E5: MLL WPE	III.E, Table XVII	60% WPE MLL assumed; demonstrated max 2–10%	Corrupts full power budget	Fixed: CW+TFLN+compressor, 40% WPE
E6: CMOS backend	IV.J.1.a (new)	Single pipeline cannot run at 100 GHz	Undefined bottleneck at 100 GHz	Fixed: 21 parallel pipelines at 4.76 GHz

All six errors have clean, engineering-sound resolutions requiring no exotic technology. The corrected architecture runs at ~100 GHz optically (3× faster), carries a higher detection margin, a physically consistent power budget, and a more manufacturable laser source.

2. Error 1 — Timing Model

Section IV.J.1, Equations 247–250 · Most Fundamental

2.1 The Paper's Claim

The paper derives the 33 GHz operating frequency as:

$$T_{\text{cycle}} = t_{\text{EAM}} + t_{\text{PD}} + t_{\text{wire}} + t_{\text{guard}} = 10 + 0.60 + 1.33 + 18.6 \approx 30.5 \text{ ps} \rightarrow 33 \text{ GHz}$$

2.2 The Physics Contradiction

■ Fatal Logic Error: Serial Summation Applied to a Wave-Pipelined System

The equation treats every latency term as a serial throughput bottleneck. This is correct only for a single-stage pipeline with no overlap. JANUS is, by construction, a deeply wave-pipelined architecture — each component processes a different pulse simultaneously. Summing their latencies gives end-to-end single-computation latency, **not** the throughput interval between successive pulses.

Component-by-component verdict:

t_{EAM} = 10 ps — EAM modulation bandwidth. This IS a throughput bottleneck because each new pulse cannot enter the modulator until the previous modulation event has completed. It correctly limits throughput to 1/10 ps = 100 GHz.

t_{PD} = 0.60 ps — Ge-fin photodetector response (265 GHz bandwidth). The detector fully recovers (0.60 ps) before the next pulse arrives at 10 ps intervals. It contributes zero to the throughput interval — it only adds to single-computation latency.

t_{wire} = 1.33 ps — On-chip electrical wire propagation. This is a latency, not a bottleneck. While pulse #1 is traversing the wire, pulse #2 enters the modulator. Multiple pulses coexist in the pipeline simultaneously.

t_{guard} = 18.6 ps for PCM routing state stabilization — This is the most damaging term. PCM switches are non-volatile. Once programmed for an inference epoch (~0.1 ms), they are passive glass. They require settling only after a weight-loading event. Adding 18.6 ps of settling overhead to every 30.5 ps clock cycle is physically nonsensical. This single spurious term accounts for 61% of the cycle time.

2.3 Correct Analysis

The true throughput-limiting analysis:

Component	Role	Value	Throughput Bottleneck?
EAM modulation	Modulates each new pulse	10 ps → 100 GHz	YES — limits throughput
Ge-fin detector	Recovers between pulses	0.60 ps → 265 GHz	NO — not limiting
Wire propagation	Latency only, pipelined	1.33 ps (latency)	NO — multiple pulses coexist
Beneš 15-stage routing	750 ps total latency, pipelined	750 ps (latency only)	NO — wave-pipelined
PCM settling	Once per 0.1 ms epoch	~0.1 ms (amortised)	NO — not per-cycle
CMOS CRT adder-tree	210 ps latency, pipelined	210 ps (latency only)	NO — runs in parallel

The correct throughput-limiting equation is:

```
f_clk = 1 / max(t_EAM, t_detector_recovery, t_CMOS_latch_setup) = 1 / 10 ps = 100 GHz (EAM-limited)
```

The end-to-end single-computation latency (not the clock period) is:

```
T_latency = t_Benes_routing + t_CRT_reconstruction = 750 ps + 210 ps = 960 ps (31 pipeline stages at 33 GHz, 10 at 100 GHz)
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✓ Corrected Timing Model

Operating frequency: ~100 GHz (limited by EAM bandwidth, not PCM settling). The architecture is wave-pipelined: the clock period equals the EAM modulation time (10 ps), while the Beneš routing fabric (750 ps) and CRT backend (210 ps) operate as deep pipelines with multiple computations in flight simultaneously.

Note on the 33 GHz figure: The paper's stated 33 GHz may coincidentally approximate the CMOS backend's sustainable throughput ceiling (4–6 GHz at TSMC N3 for a single pipeline, which must be resolved via parallelism — see Section 7). However, the stated physics derivation is wrong, and the optical system itself supports 100 GHz.

2.4 Throughput Impact

Configuration	Original (33 GHz, 64-bit)	Corrected (100 GHz, 64-bit)	Multiplier
Mini (32 tiles)	57.6 TMAC/s	~174.5 TMAC/s	~3.0×
Edge (32 tiles)	229.8 TMAC/s	~696.5 TMAC/s	~3.0×
Datacenter (32 tiles)	919.2 TMAC/s	~2,785 TMAC/s	~3.0×

All throughput figures represent exact integer 64-bit operations. The 3× gain is purely from correcting the timing model; no architectural change is required.

3. Error 2 — GaP Microring Spectral Mismatch

Section III.D.4 · GaP microring Q-factor vs. pulse bandwidth

3.1 The Paper's Claim

Section III.D.4 specifies GaP microring resonators with $Q > 10,000$ (radius $\approx 20\text{ }\mu\text{m}$) for phase-sensitive amplification. The signal pulses are 1.5 ps wide, generated by a mode-locked laser at 33 GHz.

3.2 The Physics Contradiction

■ Spectral Mismatch: Pulse 15× Wider Than Cavity Passband

Cavity bandwidth at $Q = 10,000$, $\lambda = 1550\text{ nm}$ ($f_{\text{c}} \approx 193\text{ THz}$):

$$\Delta f_{\text{cavity}} = f_{\text{c}} / Q = 193\text{ THz} / 10,000 = 19.3\text{ GHz}$$

Transform-limited Gaussian pulse bandwidth at 1.5 ps:

$$\Delta f_{\text{pulse}} = 0.44 / \tau = 0.44 / 1.5\text{ ps} \approx 293\text{ GHz}$$

The pulse is **~15× wider spectrally** than the cavity passband. The ring acts as a brutal spectral filter, rejecting the overwhelming majority of the pulse's energy and temporally stretching the transmitted remnant. This breaks both the power budget and the timing closure simultaneously. This is an unambiguous, closed-form physics error.

3.3 The Correct Fix

Two parameters must change together to achieve spectral matching:

- Pulse width: 1.5 ps $\rightarrow$ 5 ps
- GaP microring Q-factor: $> 10,000 \rightarrow \approx 2,000$

Verification:

$$\Delta f_{\text{pulse}} (5\text{ ps}) = 0.44 / 5\text{ ps} \approx 88\text{ GHz} \quad \Delta f_{\text{cavity}} (Q=2000) = 193\text{ THz} / 2,000 \approx 96.5\text{ GHz}$$

The 96.5 GHz cavity now accommodates the 88 GHz pulse with $\approx 9\%$ margin. The spectral mismatch is fully resolved.

✓ Corrected GaP Microring Parameters

Pulse width: 5 ps (down from 1.5 ps)

GaP ring Q-factor: $\sim 2,000$ (down from $>10,000$)

Pump power impact: Q drops $5\times \rightarrow$ circulating power requirement rises $5\times$. However, with cavity buildup $B = Q/\pi \approx 637$ at $Q=2,000$, the external pump remains in the single-digit μW per site range — thermally and electrically negligible.

Secondary benefit: At 100 GHz clock with 5 ps pulses, a CW laser driven through a TFLN MZM at 100 GHz naturally produces sinusoidal modulation that a passive chirped Bragg grating compresses into ~ 5 ps pulses. This also resolves Error 5 (see Section 6).

3.4 Spectral Budget Comparison

Parameter	Original (wrong)	Corrected
Pulse width	1.5 ps	5 ps

Pulse bandwidth ($0.44/\tau$)	~293 GHz	~88 GHz
GaP ring Q-factor	>10,000	~2,000
Cavity passband (f_{cav}/Q)	~19.3 GHz	~96.5 GHz
Spectral match ratio	15× mismatch (broken)	≈1.1× (matched with margin)
Pump external power / site	~1.3 μW	~6.5 μW (negligible increase)

Section III.B.5 narrative vs. Table VIII

Section III.B.5 explicitly states: "P_{signal} ≈ -7.5 dBm" at the Ge-fin detector, leading to a detection margin of (-7.5) - (-31.76) = +24.26 dB.

Tracing the optical chain defined by the paper itself:

The architecture actually has a 10 dB larger margin than stated. This is favorable — it increases robustness to fabrication loss variations — but must be corrected in the narrative before external review.

5. Error 4 — Laser Power Budget Arithmetic

Section III.E.3 and Table XVII

5.1 The Paper's Claim

Section III.E.3 (p.16) states: "each laser delivering a time-averaged power of approximately 1.6 mW (−20 dBm peak, 5% duty cycle, and split loss accounted)." Table XVII then uses $1.6 \text{ mW} \times 8,192 \text{ lasers} / 60\% \text{ WPE} = 13.33 \text{ W}$ for the datacenter laser budget.

5.2 The Arithmetic Contradiction

■ Factor-of-3,200 Error in Time-Averaged Power

At −20 dBm peak = 0.01 mW and 5% duty cycle:

Time-average = $0.01 \text{ mW} \times 0.05 = 0.0005 \text{ mW}$ (0.5 μW) Paper claims: 1.6 mW — a factor of 3,200 larger.

The 1.6 mW figure appears to have been computed through an inconsistent route: working backward from Table XVII's total laser power and dividing by port count without correctly accounting for duty cycle. Since Table XVII's 13.33 W laser budget is derived from this 1.6 mW figure, the entire laser power row in Table XVII is approximately 50× too large.

5.3 Correct Derivation

Working forward from the detector requirement:

Required at detector (PSA input → GaP front-end → Beneš): Detector target: +2.5 dBm (corrected, §4) Beneš routing loss: +7.5 dB ($15 \times 0.5 \text{ dB}$) GaP PSA gain: −30 dB (undoing gain) 1:64 passive split: +18.06 dB

Required laser peak: $2.5 + 7.5 - 30 + 18.06 \approx -1.94 \text{ dBm} \approx 0.64 \text{ mW}$ peak Time-averaged laser power = $0.64 \text{ mW} \times 0.05$ (duty cycle) = 0.032 mW per laser Total datacenter laser power: $0.032 \text{ mW} \times 8,192 \text{ lasers} / 0.40 \text{ WPE (CW+TFLN, see §6)} \approx 0.66 \text{ W}$ (Using original 60% WPE MLL assumption: $0.032 \text{ mW} \times 8,192 / 0.60 \approx 0.44 \text{ W}$)

✓ Corrected Laser Power Budget

Per-laser time-averaged optical power: 0.032 mW (not 1.6 mW)

Total datacenter laser electrical power: ~0.44–0.66 W (not 13.33 W)

This is a reduction of approximately 20–30× from the stated figure. The laser row in Table XVII must be revised. Total system power for the Datacenter configuration drops from 120–140 W to approximately 107–127 W once this correction is applied (other rows are unaffected).

Section III.E and Table XVII · 60% WPE assumed

Table XVII lists the laser source as 'Laser Source (60% WPE)'. This 60% wall-plug efficiency figure is applied to a mode-locked pulsed laser (MLL) running at 33 GHz repetition rate with 1.5 ps pulses.

■ 60% WPE is Demonstrated Only for CW Lasers — Not MLLs

- State-of-the-art MLLs at 10–100 GHz repetition rates: 2–10% WPE demonstrated
- Passive MLLs: typically 1–5% WPE
- Active MLLs with RF drive: 3–8% WPE
- 60% WPE is physically unreachable for any pulsed GHz laser today

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6.4 Demonstrated References for CW+TFLN Solution

TFLN modulator >100 GHz BW: Wang et al. (2018), Nature, doi:10.1038/s41586-018-0551-y — <https://doi.org/10.1038/s41586-018-0551-y>

TFLN MZM with $V_{\pi} < 1V$: He et al. (2019), Nature Photonics, doi:10.1038/s41566-019-0378-6 — <https://doi.org/10.1038/s41566-019-0378-6>

InP-on-Si CW laser 40% WPE review: Coldren et al., IEEE J. Sel. Top. Quantum Electron. — <https://ieeexplore.ieee.org/document/6175023>

7. Error 6 — CMOS Backend Bottleneck at 100 GHz

New issue introduced by the 100 GHz upgrade · Resolved by parallel pipelining

7.1 The Problem

Once the optical pipeline is correctly clocked at 100 GHz, optical results arrive every 10 ps. The CRT reconstruction pipeline has approximately 210 ps total latency. A single serial CMOS pipeline at TSMC N3 cannot close timing at 100 GHz — complex adder trees operate at 4–6 GHz on N3 even under maximum voltage stress.

■ Single-Pipeline CMOS Cannot Match 100 GHz Optical Throughput

The paper's Section IV.J.1.a describes the CMOS pipeline as a single deep-pipelined structure with adder-tree latency ~210 ps. At 33 GHz this implied ~7 pipeline stages — feasible. At 100 GHz, a single pipeline running at 100 GHz would require a 10 ps clock cycle for complex arithmetic logic, which no CMOS process node can achieve for multi-bit adder trees.

7.2 The Industry-Standard Solution: Time-Interleaved Parallel Pipelines

The principle is fundamental to all high-speed digital-optical interfaces: the pipeline does not need to run at 100 GHz. It only needs to produce results at a 100 GHz rate. These are different requirements.

Required number of parallel CRT engines:

$N_{\text{pipelines}} = t_{\text{CRT}} / t_{\text{optical}} = 210 \text{ ps} / 10 \text{ ps} = 21$ parallel pipelines
Each pipeline runs at: $f_{\text{CMOS}} = 100 \text{ GHz} / 21 \approx 4.76 \text{ GHz}$ (comfortable at TSMC N3)

7.3 Why This is Standard Industry Practice

Application	Technique	Parallelism Factor	Reference
High-speed ADCs (>10 GSps)	Time-interleaved ADC cores	4–16×	Texas Instruments ADC12DJ5200RF, Analog Devices AD9213
100G/400G Ethernet SerDes	1:N DMUX after TIA	4–8×	All 400G QSFP-DD transceivers (Inphi, Marvell, Broadcom)
HBM memory interface	Wide parallel buses at lower f	128–1024×	HBM2E, HBM3 standards (JEDEC)
Apple M-series die-to-die	Parallel slow lanes	Many	Apple Silicon architecture (WWDC 2020)
DSP polyphase filter banks	Polyphase decomposition	N×	Vaidyanathan, 'Multirate Systems and Filter Banks'

7.4 Power Advantage of Parallel Low-Voltage Pipelines

Dynamic CMOS power: $P = \alpha \times C \times V^2 \times f$

At TSMC N3: 4.76 GHz operation: $V_{\text{DD}} \approx 0.65 \text{ V}$ (comfortable, near-threshold)
Hypothetical 100 GHz: $V_{\text{DD}} > 1.0 \text{ V}$ (impossible for complex logic)
Power ratio (single fast vs. 21 parallel slow): $P_{\text{fast}} / P_{21\text{slow}} = (1.0^2 \times 100) / (21 \times 0.65^2 \times 4.76) = 100 / 42.2 \approx 2.37\times$
Parallel approach uses roughly LESS THAN HALF the dynamic power of a hypothetical single fast pipeline, at identical throughput.

7.5 JANUS Integration Architecture

The physical integration follows naturally from the existing JANUS I/O stack:

Ge-fin detector (265 GHz bandwidth)

- TIA (already in Table XVII I/O Interface)
- 1:21 high-speed electronic DMUX (SiGe BiCMOS or InP HBT, demonstrated at 100 GHz)
- 21 parallel CRT reconstruction pipelines at ~4.76 GHz (TSMC N3, easy timing closure)
- Round-robin output MUX back to 100 GHz result stream

✓ Corrected CMOS Backend

21 parallel CRT reconstruction pipelines, each running at ~4.76 GHz

Time-interleaved: computation ID tags route each result to the correct engine

Power advantage: ~2.4× lower dynamic power vs. hypothetical single fast pipeline

Area: 21 CRT engines at TSMC N3 (~170 MTr/mm²) fits comfortably in existing CMOS footprint

1:21 DMUX at 100 GHz input: SiGe BiCMOS front-end, identical to 100G/400G transceiver chipsets

7.6 Sequencing Detail

CRT reconstruction requires residues from all k optical tiles for the same computation to arrive at the same pipeline instance. With 21 parallel pipelines, results are tagged with a computation ID by the CMOS encoder front-end. The DMUX routes by ID mod 21. Since all k residue tiles for one computation are clocked synchronously (shared optical master clock), they arrive within the same 10 ps window and are routed together. No out-of-order reconstruction problem arises.

7.7 References for 100 GHz DMUX Technology

InP HBT 100 Gbit/s DMUX: Möller et al. (2003), Electronics Letters — <https://ieeexplore.ieee.org/document/1244741>

SiGe BiCMOS 100G+ receivers: Lischke et al. (2021), Nature Photonics (same paper cited in JANUS for 265 GHz detector) — <https://doi.org/10.1038/s41566-021-00893-w>

Time-interleaved ADC survey: Vogel (2005), IEEE Trans. Circuits Syst. — <https://ieeexplore.ieee.org/document/1410065>

8. Corrected Architecture Summary

Full parameter comparison: original paper vs. all corrections applied

8.1 Parameter Comparison Table

Parameter	Original Paper	Corrected Value	Change Source
Operating frequency	33 GHz	~100 GHz (optical)	Error 1: timing model
Timing model	Serial sum: $t_{\text{EAM}} + t_{\text{PD}} + t_{\text{wire}} + t_{\text{guard}}$	Wave-pipelined, EAM-limited	Error 1
Guard band term (t_{guard})	18.6 ps (per cycle)	0 (amortised over epoch)	Error 1
Pulse width	1.5 ps	5 ps	Error 2: spectral match
GaP microring Q	> 10,000	~2,000	Error 2: spectral match
GaP cavity bandwidth	~19.3 GHz	~96.5 GHz	Error 2
Spectral ratio pulse/cavity	~15× (mismatched)	~0.91× (matched)	Error 2
Detector signal power	−7.5 dBm (narrative)	+2.5 dBm (correct)	Error 3
Detection margin	+24.26 dB	+34.26 dB	Error 3
Per-laser time-avg power	1.6 mW	0.032 mW	Error 4: factor-3200
Datacenter laser electrical	13.33 W	~0.44–0.66 W	Error 4
Laser source type	MLL at 33 GHz	CW + TFLN MZM + Bragg grating	Error 5
Laser WPE	60% (impossible for MLL)	35–40% (demonstrated CW)	Error 5
CMOS backend pipelines	1 (bottleneck undefined)	21 parallel at ~4.76 GHz	Error 6
CMOS clock per pipeline	Undefined / implied 33 GHz	~4.76 GHz (comfortable N3)	Error 6
DC throughput (64-bit exact)	919.2 TMAC/s	~2,785 TMAC/s (optical ceiling)	Error 1
Total system power (DC)	120–140 W	~107–127 W (laser corrected)	Errors 4+5
CMOS dynamic power	Baseline	~2.4× lower (voltage scaling)	Error 6

8.2 What Remains Unchanged

All six errors are at the execution-level physics and power budget. The core architectural innovations are unaffected and remain valid:

- One-Hot spatial encoding: eliminates analog precision collapse by construction
- Non-volatile PCM Beneš routing: eliminates ~10 MW thermo-optic static power
- Single-wavelength operation: structurally suppresses FWM/XPM nonlinear interference
- CRT-based fault visibility: converts silent corruption into explosive arithmetic invalidity
- Z-axis thermal partitioning: decouples CMOS heat from photonic stability requirements

- Spatial RNS decomposition: scales precision through parallelism, not analog fidelity

9. Open Problems Registry

Remaining unresolved challenges after all corrections applied

9.1 Pre-Existing Open Problems (Appendix F, Janus.pdf)

OP-1: Five-stratum SiPh yield characterisation. Three- and two-stratum SiPh platforms are demonstrated. Extending to five strata with ~203M PCM switches per layer at acceptable interlayer optical loss and wafer-scale yield has not been characterised. Critical questions: cumulative interlayer insertion loss across five strata, and yield of vertical optical vias at required density.

OP-2: Distributed degenerate GaP PSA pump phase-locking at scale. Degenerate PSA in individual waveguides is demonstrated. The open problem is maintaining phase coherence of the degenerate pump across millions of spatially separated GaP zones spanning a multi-tile datacenter fabric. The paper describes four passive mitigation strategies (GaP routing isolation, segmented island growth, global vertical pump illumination, 100 μm scale limit), but formal experimental characterisation of residual phase error budget across a full multi-tile fabric remains future work.

OP-3: Formal multi-fault CRT detectability bounds. CRT provides strong visibility under single-residue faults. A formal adversarial model for simultaneous multi-tile faults, correlated failures (shared thermal events across adjacent tiles), and faults confined within a single residue domain (not crossing CRT inconsistency boundary) has not been established. Closed-form detectability bounds under worst-case conditions remain open.

9.2 New Open Problem Introduced by 100 GHz Upgrade

OP-4: CMOS backend throughput verification at 100 GHz input rate. The parallel pipeline solution (21 engines at 4.76 GHz) is the correct engineering approach, but the CMOS adder-tree depth at TSMC N3 under the specific CRT reconstruction algorithm must be formally verified to confirm 4.76 GHz timing closure. The CRT engine complexity scales with the number of residue tiles k and the modulus set bitwidths. A gate-level synthesis pass (using tools such as Synopsys Design Compiler with N3 PDK) is required before finalising the 21-pipeline claim with precision.

9.3 Priority Ordering

Priority	Open Problem	Blocking?	Next Step
1 (Highest)	OP-2: PSA pump phase-locking	Yes, for datacenter scale	On-chip phase distribution architecture design
2	OP-4: CMOS 4.76 GHz closure	Yes, for 100 GHz claim	Gate-level synthesis with N3 PDK
3	OP-1: 5-stratum yield	Yes, for fabrication	Partner with IMEC / GlobalFoundries SiPh PDK
4 (Lower)	OP-3: Multi-fault CRT bounds	No (safety margin, not blocking)	Formal fault model analysis

10. References and Verified Links

All sources directly relevant to the errors and their solutions

10.1 Error 1 — Timing / Wave-Pipelining

Ge-fin photodetector 265 GHz bandwidth

Lischke et al. (2021), Nature Photonics, vol. 15, pp. 925–931

<https://doi.org/10.1038/s41566-021-00893-w>

EAM bandwidth at 100 GHz (Ge EAM on Si)

Steckler et al. (2024), IEEE Trans. Electron Devices, vol. 71, pp. 3417–3423

<https://ieeexplore.ieee.org/document/10540259>

Wave-pipelining in photonic circuits (textbook reference)

Saleh & Teich, Fundamentals of Photonics, 3rd ed., Wiley, 2019 — Ch. 22

<https://www.wiley.com/en-us/Fundamentals+of+Photonics%2C+3rd+Edition-p-9781119506874>

10.2 Error 2 — GaP Microring Spectral Match

GaP-on-Si nonlinear photonics (nonlinear coefficient, Q-factor)

Wilson et al. (2020), Nature Photonics, vol. 14, pp. 57–62 [JANUS Ref. 4]

<https://doi.org/10.1038/s41566-019-0537-9>

Transform-limited pulse bandwidth formula ($0.44/\tau$ for Gaussian pulses)

Saleh & Teich, Fundamentals of Photonics, 3rd ed., Wiley, 2019 — §3.1

<https://www.wiley.com/en-us/Fundamentals+of+Photonics%2C+3rd+Edition-p-9781119506874>

GaP integrated optical parametric amplifier: 31.5 dB gain at 8.5 mW pump

Xiao et al. (2024), OFC 2024, paper Th1D.4 [JANUS Ref. 34]

<https://opg.optica.org/abstract.cfm?uri=OFC-2024-Th1D.4>

Chirped fiber Bragg grating pulse compression (passive, < 2 dB loss)

Kashyap, Fiber Bragg Gratings, 2nd ed., Academic Press, 2009 — Ch. 8

<https://doi.org/10.1016/B978-0-12-372579-0.X0001-6>

10.3 Errors 3 & 4 — Power Budget and Detector Margin

Optical power budget methodology for silicon photonic links

Miller (2017), J. Lightwave Technol., vol. 35, no. 3, pp. 346–396 [JANUS Ref. 11]

<https://doi.org/10.1109/JLT.2017.2647779>

Transimpedance amplifier noise floor analysis for photonic receivers

Winzer & Essiambre (2006), J. Lightwave Technol., vol. 24, no. 12 [JANUS Ref. 21]

<https://ieeexplore.ieee.org/document/4014503>

10.4 Error 5 — Laser Source (CW + TFLN Solution)

TFLN MZM >100 GHz bandwidth demonstrated

Wang et al. (2018), Nature, vol. 562, pp. 101–104

<https://doi.org/10.1038/s41586-018-0551-y>

TFLN MZM $V_{\pi} < 1V$ at 40 GHz

He et al. (2019), Nature Photonics, vol. 13, pp. 359–364

<https://doi.org/10.1038/s41566-019-0378-6>

TFLN photonics review: modulators, frequency combs, nonlinear optics

Zhu et al. (2021), Advanced Photonics, vol. 3, no. 2

<https://doi.org/10.1117/1.AP.3.2.028002>

CW InP-on-Si laser integration for silicon photonics

Liang & Bowers (2010), Nature Photonics, vol. 4, pp. 511–517

<https://doi.org/10.1038/nphoton.2010.167>

60% WPE CW laser source (Ren et al. 2020 — cited by JANUS for MLL, incorrect application)

Ren et al. (2020), J. Semiconductors, vol. 41, no. 3 [JANUS Ref. 35] — CW only

<https://doi.org/10.1088/1674-4926/41/3/032901>

10.5 Error 6 — CMOS Backend / Parallel Pipelines

Time-interleaved ADC architecture (canonical reference)

Vogel (2005), IEEE Trans. Circuits Syst. I, vol. 52, no. 12

<https://ieeexplore.ieee.org/document/1410065>

InP HBT 100 Gbit/s DMUX (1:2 at 100 Gbps, extendable to 1:N)

Möller et al. (2003), Electronics Letters, vol. 39, no. 1

<https://ieeexplore.ieee.org/document/1244741>

SiGe BiCMOS 100G+ optical receiver front-end

Liu et al. (2021), Microelectronics Journal, vol. 113 [JANUS Ref. 30]

<https://doi.org/10.1016/j.mejo.2021.105083>

TSMC N3 process node timing characteristics (public roadmap)

TSMC Technology Symposium 2022 — N3 performance data

<https://pr.tsmc.com/english/news/2957>

Polyphase filter bank / parallel pipeline theory (textbook)

Vaidyanathan, Multirate Systems and Filter Banks, Prentice Hall, 1992

<https://www.pearson.com/en-us/subject-catalog/p/multirate-systems-and-filter-banks/P100000218427>

400G QSFP-DD transceiver architecture (1:4 DMUX standard)

IEEE 802.3bs Standard (400GbE) — publicly available

<https://standards.ieee.org/ieee/802.3bs/6375/>

10.6 Core JANUS Architecture References (Unchanged, Confirmed Valid)

PCM non-volatile photonic switching (zero static power demonstrated)

Wuttig, Bhaskaran & Taubner (2017), Nature Photonics, vol. 11 [JANUS Ref. 6]

<https://doi.org/10.1038/nphoton.2017.126>

GST zero-static-power PCM switch in SiPh

Fang et al. (2022), Nature Nanotechnology, vol. 17 [JANUS Ref. 7]

<https://doi.org/10.1038/s41565-022-01153-w>

CRT error amplification / RNS fault visibility

Xia (2000), IEEE Signal Processing Letters, vol. 7, no. 11 [JANUS Ref. 10]

<https://ieeexplore.ieee.org/document/873549>

GaP-on-Si CMOS-compatible nonlinear photonics

Wilson et al. (2020), Nature Photonics [JANUS Ref. 4]

<https://doi.org/10.1038/s41566-019-0537-9>

3D SiPh heterogeneous integration

Skacel et al. (2021), IEEE ECTC [JANUS Ref. 39]

<https://ieeexplore.ieee.org/document/9501440>

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